

QUENTIN BACQUELÉ

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 GitHub |  Google Scholar

PhD candidate in acoustic biogeography. I believe bioacoustics should be scaled to the biogeographic level, uncovering patterns that remain invisible at local scales. The rise of machine learning in bioacoustics, combined with vast amounts of available data and techniques from spatial, phylogenetic, and functional ecology, is opening the door to a new macroecological framework for studying acoustic communication.

EDUCATION

PhD Candidate in Acoustic Communication, Biogeography & AI <i>EPHE / ENES / CEFE</i>	Saint-Étienne, France 2024 – Present
M.Sc. International Master in Bioacoustics (MoBi) <i>Jean Monnet University</i>	Saint-Étienne, France 2023 – 2024
M.Sc. Ecophysiology, Ecology, and Ethology (EEE) <i>Strasbourg University – Double Degree</i>	Strasbourg, France 2021 – 2024
M.Sc. Interdisciplinary Approach in Data Science (AISD) <i>Strasbourg University – Double Degree</i>	Strasbourg, France 2021 – 2024
University Degree in Bioacoustics <i>Jean Monnet University</i>	Saint-Étienne, France 2023
B.Sc. in Life Sciences, specializing in Ecology <i>Savoie Mont-Blanc University</i>	Chambéry, France 2020 – 2021

PUBLICATIONS

HAC-Net: Learning Natural Units from Acoustic Change Quentin Bacquéle, Jean-Yves Barnagaud, Frederic Theunissen, Nicolas Mathevon	2025 NeurIPS 2025
AI Decodes Chimpanzee Vocabulary Items Antoine Valet, Quentin Bacquéle, Cédric Girard-Buttoz, Roman M Wittig, Catherine Crockford	2025 NeurIPS 2025
A novel contour-based approach for quantifying cryptic vocal dialects Quentin Bacquéle, Barbara Freitas, Borja Milá, Christophe Thébaud, et al.	In Prep
The global biogeography of passerine songs Quentin Bacquéle, Jean-Yves Barnagaud, Cyrille Violle, et al.	In Prep

RESEARCH EXPERIENCE

A novel contour-based approach for quantifying cryptic vocal dialects <i>University of California at Berkeley – Supervised by F. Theunissen & N. Mathevon</i>	Berkeley, USA 2024
Can chimpanzees flexibly combine calls to create new meanings? <i>ENES Lab, Jean Monnet University – Supervised by F. Levrero, C. Crockford & C. Girard-Buttoz</i>	Saint-Étienne, France 2023
An AI-based approach for planarian behavior analysis <i>Institute of Cellular and Integrative Neuroscience (CNRS) – Supervised by H. Cadiou</i>	Strasbourg, France 2023
Passive Acoustic Monitoring of wild bottlenose dolphins <i>Bottlenose Research Institute (BDRI) – Supervised by S. Methion & B. Diaz Lopez</i>	Ría de Arousa, Spain 2022

PRESENTATIONS

HAC-Net: Learning Natural Units from Acoustic Change

NeurIPS 2025 – Poster Presentation

San Diego, USA

Dec 2025

An AI based approach for planarian behavior analysis

NeuroFrance 2023 – Poster Presentation

Lyon, France

2023

Passive Acoustic Monitoring to discover the hidden nightlife of dolphins

34th European Cetacean Society Conference – Poster Presentation

O Grove, Spain

2023

GRANTS & FELLOWSHIPS

PhD Stipend (PSL-EPHE EABIS Grant) – €80,400

University Paris-Sciences-Lettres

2024 – 2027

Mobility Grant – €500

EPHE

2024

TEACHING

Analyzing human and animal sounds (3h)

AI PSL Week

ENS ULM

2026

Sound Resynthesis (3h)

Bioacoustics Winter School

Jean Monnet University

2026

Sound Resynthesis (4h)

M.Sc. Bioacoustics

Jean Monnet University

2025

Statistics (16h)

Master 2 Ecology

Jean Monnet University

2024 – 2025

Ecology - Practical Work (28h)

B.Sc. Life Sciences (L2 & L3)

Jean Monnet University

2024

MENTORSHIP & SUPERVISION

Python for Bioacoustics

Mentorship of 2 students

UC Berkeley

2024

M.Sc. Project Co-Supervisor

M.Sc. in Bioacoustics (MoBi)

Jean Monnet University

2024

- Topic: Biogeography of rhythm in passerine birds
- Topic: Phylogeny of non-linear phenomena in mammals

OTHER EXPERIENCE

FauNet (Founder)

Client: SILABE – Facial recognition of rhesus monkeys

Strasbourg, France

2022 – 2024

Competitions

Automatic Detection of Odontocetes' Clicks (Toulon University)

Online

2023